

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF
ENGINEERING)** Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 3 – CLASS TEST (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO2	Assessment:	Assessment Tool 3 – CLASS
Weightage:	35% of CIA	Total Marks:	TEST (5 Questions × 10 Marks)
CO1 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		
Student Name:	S.Likiteswar Reddy	Reg. No.:	192425120
Section / Batch:	AI&ML	Date:	10/08/2026

Code of Conduct

I S.Likiteswar Reddy(192425120)(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: S.Likiteswar Reddy

QUESTIONS: 5 Total Marks: 50

Annexure A

Class Test Duration: 1hr

- 1.** A company developing an AI-based disease diagnosis system is deciding whether to use a traditional Machine Learning model or a Deep Learning model. Compare both approaches by discussing their feature extraction methods, data requirements, computational needs, and suitability for image-based medical diagnosis.
- 2.** A facial recognition application receives raw image data without any manually extracted features. Explain how Representation Learning enables the model to automatically learn useful features from the images and discuss its advantages over manual feature engineering.
- 3.** Two neural network models are designed for an image classification task. One model contains many hidden layers with fewer neurons, while the other has a single hidden layer with a large number of neurons. Compare these architectures and discuss how width and depth influence learning performance.
- 4.** A Deep Neural Network experiences slow learning due to the vanishing gradient problem during training. Compare the Sigmoid and Tanh activation functions and explain their impact on gradient flow, convergence speed, and overall network performance.
- 5.** An image classification model suffers from inactive neurons during training, leading to poor prediction accuracy. Evaluate the use of ReLU, Leaky ReLU, and ELU activation functions for this problem and recommend the most suitable choice with justification.
- 6.** A multiclass image classification system is designed to classify animals into several categories such as cats, dogs, birds, and horses. Explain how the Softmax activation function processes the final output layer and converts network outputs into class probabilities.
- 7.** A company collects millions of unlabeled customer behavior records and wants to discover hidden patterns without assigning labels to the data. Discuss how unsupervised training of neural networks can be applied to this problem and explain its advantages in pattern discovery.
- 8.** A music streaming platform aims to improve song recommendations by identifying hidden relationships between users and songs from historical listening data. Explain how a Restricted Boltzmann Machine (RBM) can be used in this application by describing its architecture and learning process.
- 9.** An industrial organization wants to reduce storage requirements for large sensor datasets while preserving important information. Explain how an Autoencoder performs data compression and reconstruction, and describe the role of the encoder, bottleneck layer, and decoder.

10. A smart city project plans to implement Deep Learning for applications such as face recognition, autonomous vehicles, speech recognition, and medical image analysis. Discuss how Deep Learning improves the efficiency and accuracy of these applications compared to conventional approaches.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge Course Coordinator Program Director

Signature: _____ Signature: _____ Signature: _____ Date: _____

_____ Date: _____ Date: _____

1. Traditional Machine Learning vs Deep Learning for Disease Diagnosis

Traditional Machine Learning generally requires manual feature extraction. For example, in medical images, experts may extract edges, textures, shapes, or intensity-related features before training the model.

Deep Learning automatically learns multiple levels of features directly from raw images. A CNN may learn edges in early layers, shapes in intermediate layers, and complex disease-related patterns in deeper layers.

Feature Extraction: Traditional Machine Learning uses manually designed features, whereas Deep Learning automatically learns features.

Data Requirements: Traditional Machine Learning can work well with smaller datasets, while Deep Learning usually requires large datasets.

Computational Needs: Traditional Machine Learning generally requires less computational power. Deep Learning usually needs high computational resources, often GPUs.

Image Diagnosis: Deep Learning is highly suitable for complex image-based medical diagnosis.

Conclusion: For image-based medical diagnosis, Deep Learning is generally more suitable because it can automatically learn complex features and achieve high accuracy when sufficient data and computational resources are available.

2. Representation Learning in Facial Recognition

Representation Learning allows a neural network to automatically learn useful representations or features from raw input data.

In facial recognition, the raw image is provided directly to a Deep Learning model such as a CNN. The network learns different features through its layers. Early layers may detect edges, while deeper layers learn eyes, nose, facial structure, and other complex patterns. The final representation can then be used to identify or compare faces.

Advantages over manual feature engineering:

- Automatically learns relevant features.
- Reduces human effort.
- Discovers complex patterns that are difficult to define manually.
- Adapts features according to the training objective.
- Performs well with large image datasets.

- Can provide better accuracy for complex recognition tasks.

Thus, Representation Learning makes facial recognition more flexible and scalable than manually designing facial features.

3. Width vs Depth in Neural Network Architecture

Depth refers to the number of hidden layers, while width refers to the number of neurons in each layer.

A deep and narrow network contains many hidden layers with fewer neurons per layer. It can learn hierarchical representations, where each layer transforms the representation learned by the previous layer. For image classification, the network may learn edges, textures, shapes, objects, and finally classes.

A shallow and wide network has fewer hidden layers but many neurons. It can represent complex functions, but it has fewer levels of hierarchical feature transformation.

Deep Network:

- Many hidden layers.
- Learns hierarchical features.
- Often efficient for complex patterns.
- Can model complex relationships.
- Excessive depth can make training difficult.

Wide Network:

- Few hidden layers.
- Many neurons.
- Can approximate complex functions.
- May require many parameters.

Conclusion: Increasing depth generally helps Deep Learning models learn hierarchical and complex representations, especially in image tasks. However, an appropriate balance between width and depth is important.

4. Sigmoid vs Tanh and the Vanishing Gradient Problem

Sigmoid and Tanh are activation functions commonly used in neural networks.

Sigmoid: $\sigma(x) = 1/(1+e^{-x})$. Its output ranges from 0 to 1. When the input is very large or very small, the function becomes saturated and its gradient becomes very small. This can cause the vanishing gradient problem.

Tanh: $\tanh(x) = (e^x - e^{-x}) / (e^x + e^{-x})$. Its output ranges from -1 to 1. Tanh is zero-centered and generally provides stronger gradients around zero than sigmoid, so it can often converge faster.

Comparison:

- Sigmoid output: 0 to 1.
- Tanh output: -1 to 1.
- Sigmoid is not zero-centered.

- Tanh is zero-centered.
- Both can suffer from vanishing gradients.
- Tanh generally provides better gradient flow than sigmoid.

Conclusion: Tanh is generally preferable to sigmoid for hidden layers when choosing between the two because it is zero-centered and usually provides better gradient flow. Modern deep networks commonly use ReLU-family activations to further reduce vanishing-gradient problems.

5. ReLU vs Leaky ReLU vs ELU

Inactive neurons can occur with ReLU when neurons continuously receive negative inputs. This is known as the dying ReLU problem.

ReLU: $f(x) = \max(0, x)$. It is simple and computationally efficient, but negative inputs produce zero output and can cause dead neurons.

Leaky ReLU allows a small negative value for negative inputs. This reduces the dying-neuron problem because negative inputs still have a small gradient.

ELU also provides negative outputs for negative inputs and approaches a fixed negative value. It can reduce the dead-neuron problem and provides smoother behavior for negative inputs.

Comparison:

- ReLU: simple and fast, but can cause dead neurons.
- Leaky ReLU: keeps a negative gradient and reduces inactive neurons.
- ELU: provides a smoother negative region but is more computationally expensive.

Recommendation: Leaky ReLU is a strong choice when inactive neurons are the main problem because it retains a small gradient for negative inputs while remaining computationally simple. ELU is also a good alternative when smoother negative activation is beneficial.

6. Softmax for Multiclass Image Classification

Softmax is commonly used in the output layer of a multiclass neural network.

Suppose the model classifies an image into cats, dogs, birds, and horses. The final neural network layer produces raw scores called logits. Softmax converts these scores into probabilities using:

$$P(y=i) = e^{(z_i)} / \sum e^{(z_j)}$$

The resulting probabilities are between 0 and 1, and all class probabilities add up to 1.

For example:

Cat = 0.10
Dog = 0.70
Bird = 0.05
Horse = 0.15

The highest probability is Dog (70%), so the model predicts that the image belongs to the dog class.

Thus, Softmax converts the final network outputs into an interpretable probability distribution across

multiple classes.

7. Unsupervised Training for Customer Behavior Analysis

Unsupervised learning is useful when a company has large amounts of unlabeled data. Millions of customer behavior records can be given to a neural network without predefined labels. The network attempts to discover hidden structures and patterns.

Methods such as Autoencoders, Restricted Boltzmann Machines, self-organizing maps, and deep clustering methods can be used.

Possible discoveries include:

- Similar purchasing behavior.
- Customer segments.
- Frequently occurring behavior patterns.
- Unusual or anomalous behavior.
- Hidden relationships between activities.

Advantages:

1. Does not require manually labeled data.
2. Can process very large datasets.
3. Discovers previously unknown patterns.
4. Helps with customer segmentation.
5. Can identify anomalies.
6. Reduces the cost and effort of manual labeling.

Therefore, unsupervised neural networks are particularly useful when labeled customer data is unavailable or expensive to obtain.

8. Restricted Boltzmann Machine for Music Recommendation

A Restricted Boltzmann Machine (RBM) is a generative neural network that can learn hidden relationships in data.

An RBM contains two layers: a visible layer and a hidden layer. There are no connections between neurons within the same layer.

For a music recommendation system:

- Visible units can represent users' song preferences.
- Hidden units learn hidden characteristics or preferences.
- Connections between visible and hidden units represent relationships.

The RBM is trained using historical listening data. It learns which songs tend to occur together and which hidden preferences explain users' listening behavior.

After training, the system can reconstruct or predict songs that a user may be interested in. For example, if a user frequently listens to rock and alternative music, hidden units may learn a corresponding preference pattern and recommend other related songs.

Benefits:

- Learns hidden user preferences.
- Handles large user-item datasets.
- Discovers relationships that are not explicitly provided.
- Supports personalized recommendations.

Thus, RBMs can learn latent relationships between users and songs and use them for recommendation.

9. Autoencoder for Data Compression

An Autoencoder is a neural network designed to learn a compact representation of input data and reconstruct the original data.

Its basic structure is:

Input → Encoder → Bottleneck → Decoder → Reconstruction

Encoder: The encoder takes the original sensor data and transforms it into a smaller representation.

Bottleneck Layer: The bottleneck contains a compressed representation of the most important information. For example, 100 sensor values could be represented using only 10 encoded values. The model is forced to preserve the most useful information while reducing dimensionality.

Decoder: The decoder takes the compressed representation and attempts to reconstruct the original data.

The model is trained to minimize the difference between the original input x and reconstructed output $\hat{x}$ often using a reconstruction loss such as mean squared error.

Benefits:

- Reduces storage requirements.
- Removes some noise and redundancy.
- Learns useful representations automatically.
- Reconstructs important information from compressed data.

Therefore, the bottleneck is the key component that allows the Autoencoder to perform effective data compression.

10. Deep Learning in Smart City Applications

Deep Learning uses multiple neural network layers to automatically learn complex patterns from large datasets. It can improve several smart-city applications.

Face Recognition: Deep Learning models, particularly CNN-based systems, automatically learn facial features from images and can improve recognition performance.

Autonomous Vehicles: Deep Learning can process camera and sensor information to detect vehicles, pedestrians, traffic signs, road markings, and obstacles. This helps autonomous vehicles understand their surroundings.

Speech Recognition: Deep Learning models learn relationships between audio signals and words. They can convert spoken language into text and support voice-controlled systems.

Medical Image Analysis: Deep Learning can analyze X-rays, CT scans, MRI images, and other medical images to identify patterns associated with diseases.

Compared with conventional approaches, Deep Learning:

- Automatically learns features instead of requiring extensive manual feature engineering.
- Models complex patterns effectively.
- Scales well with large datasets.
- Works especially well with images, audio, video, and sensor data.

Conclusion: Deep Learning improves smart-city applications by automatically learning complex representations from large datasets. Its ability to process images, audio, video, and sensor data makes it highly suitable for face recognition, autonomous driving, speech recognition, and medical image analysis.